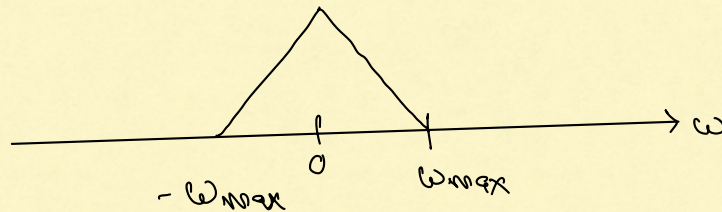


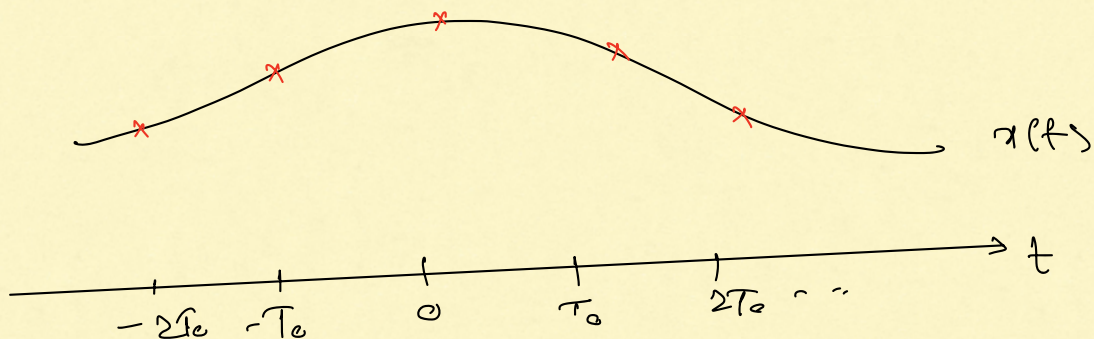
I. Sampling and Reconstruction

Recall: If $x(t)$ is a band limited signal of bandwidth ω_{max} , i.e.,

$|x(\omega)|$ low pass signal



$x(\omega) = 0$ for all $\omega \geq \omega_{max}$ and
all $\omega \leq -\omega_{max}$.

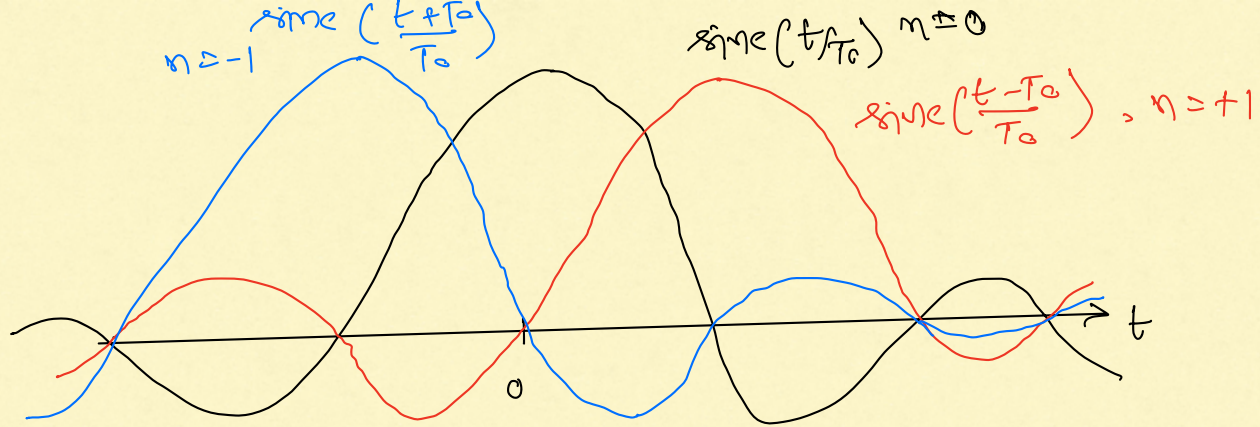


$T_0 > 0$: Called sampling interval or sampling period.

$\omega_s = \frac{2\pi}{T_0}$: sampling rate or frequency.

Sampling : $x(t) \times \sum_n \delta(t - nT_0)$

Reconstruction : $\sum_{n=-\infty}^{+\infty} x(nT_0) \cdot \text{sinc}\left(\frac{t - nT_0}{T_0}\right)$



Fact: (Practice Set 8):

For $m \neq n$:

$$\int_{-\infty}^{+\infty} \text{sinc}\left(\frac{t-nT_0}{T_0}\right) \cdot \text{sinc}\left(\frac{t-mT_0}{T_0}\right) dt = 0.$$

$\therefore$ We say that the family of signals

$$\text{sinc}\left(\frac{t-nT_0}{T_0}\right), \quad n = 0, \pm 1, \pm 2, \dots$$

are pairwise orthogonal.

— Advanced —

Precise statement of Sampling Theorem:

Theorem: If $\omega_{\max} < \frac{\omega_0}{2}$ and $\int |x(t)| dt < \infty$ and

$\int |x(t)|^2 dt < \infty$ then:

$$x(t) = \sum_{n=-\infty}^{+\infty} x(nT_0) \cdot \text{sinc}\left(\frac{t-nT_0}{T_0}\right),$$

for all t .

II. Poisson Summation Formula.

Recall: ① $\sum_n \delta(t - nT_0) \xleftrightarrow{F} \frac{2\pi}{T_0} \sum_k \delta(\omega - k\omega_0), \quad \omega_0 = \frac{2\pi}{T_0}$

② $\langle \hat{T}(\omega), X(\omega) \rangle = 2\pi \langle T(t), x(-t) \rangle.$

Choose $T(t) = \sum_n \delta(t - nT_0).$

Then: LHS of ② = $\langle \frac{2\pi}{T_0} \sum_k \delta(\omega - k\omega_0), X(\omega) \rangle$

$$= \frac{2\pi}{T_0} \sum_k \langle \delta(\omega - k\omega_0), X(\omega) \rangle$$

$$= \frac{2\pi}{T_0} \sum_k X(k\omega_0).$$

RHS of ② = $2\pi \langle \sum_n \delta(t - nT_0), x(-t) \rangle$

$$= 2\pi \sum_n \langle \delta(t - nT_0), x(-t) \rangle$$

$$= 2\pi \sum_n x(-nT_0)$$

$$= 2\pi \sum_n x(nT_0).$$

$$\Rightarrow \sum_n x(nT_0) = \sum_k \frac{1}{T_0} X(k\omega_0) \quad \text{for any}$$

Schwartz signal $x(t)$ and any $T_0 > 0$ with $\omega_0 = 2\pi/T_0$

Proof of the Poisson Summation Formula:

Idea: Start with any Schwartz signal $x(t)$.

Periodic: $y(t) = x(t) * \sum_n \delta(t - nT_0)$.

Notice: y is T_0 -periodic
 $\downarrow$

Fourier series.

Compute $y(t)$ at $t=0$ via two different methods.

$$\begin{aligned} \text{(a)} \quad y(t) &= \sum_n x(t) * \delta(t - nT_0) \\ &= \sum_n x(t - nT_0) \end{aligned}$$

$\therefore$ value of $y(t)$ at $t=0$:

$$y(0) = \sum_n x(-nT_0)$$

$$= \sum_n x(nT_0).$$

(b) observe that $y(t)$ is T_0 -periodic.

$\therefore y$ has a Fourier series representation.

$$y \overset{\text{F.S.}}{\longleftrightarrow} c_k.$$

$$C_k = \frac{1}{T_0} \int_0^{T_0} y(t) \cdot e^{-jk\omega_0 t} dt.$$

$$= \frac{1}{T_0} \int_0^{T_0} \sum_n x(t - nT_0) e^{-jk\omega_0 t} dt.$$

$$= \frac{1}{T_0} \sum_n \int_0^{T_0} x(t - nT_0) e^{-jk\omega_0 t} dt.$$

(interchanging $\sum$, $\int$ is allowed at this point

since x is Schwartz) Use $\tau = t - nT_0$:

$$\rightarrow = \frac{1}{T_0} \sum_n \int_{-nT_0}^{-nT_0+T_0} x(\tau) e^{-jk\omega_0 (\tau + nT_0)} d\tau$$

$$= \frac{1}{T_0} \sum_n \int_{-nT_0}^{-nT_0+T_0} x(\tau) \cdot e^{-jk\omega_0 \tau} \cdot e^{-jk\omega_0 nT_0} d\tau$$

1 since $\omega_0 T_0 = 2\pi$

$$= \frac{1}{T_0} \sum_n \int_{-nT_0}^{-(n-1)T_0} x(\tau) e^{-jk\omega_0 \tau} d\tau$$

$$= \frac{1}{T_0} \int_{-\infty}^{\infty} x(\tau) \cdot e^{-jk\omega_0 \tau} d\tau$$

$$= \frac{1}{T_0} \times X(k\omega_0)$$

$$\therefore c_k \text{ (of } y(t)) = \frac{1}{T_0} X(k\omega_0).$$

Since, $y(t)$ is periodized version of $x(t)$, it can be shown that the Fourier series of y converges everywhere, i.e.,

$$y(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_0 t} \text{ for all } t$$

$$\therefore y(0) = \sum_k c_k = \sum_k \frac{1}{T_0} X(k\omega_0).$$

$$\therefore \sum_n x(nT_0) = \frac{1}{T_0} \sum_k X(k\omega_0).$$

Note: The Poisson summation formula is a restatement of the result:

$$\sum_n \delta(t - nT_0) \xleftrightarrow{F} \sum_k \frac{2\pi}{T_0} \delta(\omega - k\omega_0).$$

Hence, what we have above is a proof of this result as well.